

TURAN ORUJLU

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EDUCATION

International Max Planck Research School for Intelligent Systems *December 2022 - Present*

Doctor of Philosophy Candidate

Supervisors: Charley M. Wu and Martin V. Butz

University of Tübingen, Tübingen, Germany

November 2020 - September 2022

Master of Science, Neural Information Processing

University of Osnabrück, Osnabrück, Germany

April 2016 - November 2020

Bachelor of Science, Cognitive Science

University of Pennsylvania, Philadelphia, PA, USA

September 2010 - December 2013

Bachelor of Arts, Mathematical Economics

Minor: Chemistry

RESEARCH INTERESTS

Causality, Model-based Reinforcement Learning, Intuitive Physics, Mechanistic Interpretability

POSTERS AND PRESENTATIONS

Reframing attention as a reinforcement learning problem for causal discovery

- Causal RL workshop at the RLC 2025 (oral) *Edmonton, AB, Canada. August 2025*

Intuitive Physics through the Lens of Pearl's Causal Hierarchy

- Machine Learning in Science Conference *Tübingen, Germany. July 2024*

VividDreamer: Tokenized world model with stochastic attention

- Machine Learning in Science Conference *Tübingen, Germany. July 2023*
- Analytical Connectionism Summer School *London, UK. August 2023*

RESEARCH EXPERIENCE

Tübingen AI Center, University of Tübingen

December 2022 - Present

Research Assistant, Human and Machine Cognition Lab

- Conducting research towards completion of the doctoral degree requirements

Max Planck Institute for Biological Cybernetics

November 2021 - November 2022

Research Assistant, Computational Neuroscience Department

- Master thesis project
- Lab rotation

Institute of Cognitive Science, University of Osnabrück

December 2016 - February 2017,

Research Assistant, Neuroinformatics Research Group

November 2018 - July 2019

- Contributed to the development of the chatbot: www.flu-prediction.com/ask-watson.

TEACHING EXPERIENCE

Tübingen AI Center, University of Tübingen

Teaching Assistant, Human and Machine Cognition Lab

April 2023 - June 2023,

November 2024 - February 2025

- Corrected weekly homework assignments accompanying the General Principles of Human and Machine Learning lecture
- Hosted weekly tutorials for the students of the General Principles of Human and Machine Learning lecture

Institute of Cognitive Science, University of Osnabrück

Teaching Assistant, Neuroinformatics Research Group

October 2017 - February 2018,

October 2018 - February 2019

- Corrected weekly homework assignments accompanying the Neuroinformatics lecture
- Hosted weekly Q&A sessions for the students of the Neuroinformatics course
- Corrected the final exam for the Neuroinformatics course

Institute of Mathematics, University of Osnabrück

Teaching Assistant, Stochastics Research Group

October 2018 - February 2019

- Corrected weekly homework assignments accompanying the Probability lecture
- Hosted weekly Tutorials for the students of the Probability course
- Corrected the final exam for the Probability course

SUPERVISION

Trong Vu Le

Master Rotation

University of Tübingen

2025

- The role of positional embeddings in transformer-based transition models for representation learning in compositional environments

Marcel De Sutter

Master Thesis

University of Tübingen

2023

- Physical and psychological reasoning in artificial cognitive systems: A benchmark study of Loci's object vs. agent recognition

AWARDS AND DISTINCTIONS

Stipend covering the Master phase of the 5-year track of the IMPRS for the Mechanisms of Mental Function and Dysfunction

DAAD-PROMOS travel grant for the research stay at the Johns Hopkins University

DAAD RISE Worldwide Scholarship for the period 08.2019-10.2019

Germany Scholarship (Deutschlandstipendium) for the academic year 2018/19

INVITED TALKS

Reframing attention as a reinforcement learning problem for causal discovery

- Computational Machinery of Cognition lab talk

TU Dresden. December 2025